

Get It Right the First Time Through Cheap and Easy DIY Usability Testing

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Drupal GovCon

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Overview

- Introduction
- What is Usability
- Who, What, Where, and When
- Free and/or Inexpensive Tools



Introduction

Disclaimer

Let's get this out of the way first!

Disclaimer

We are not Usability experts

- We'll share easy DIY techniques & tools

We won't tell you how to fix your website

- But we will point you in the right direction

Scope

What we'll cover and what we won't.

What we will cover

- What is usability?
- Why should I bother?
- How can I test my site?
- How do I keep my site usable?

What we won't cover

- Usability best practices
- Statistical analysis of results
- Every type of usability test



What is Usability?

What is usability?

“Usability testing refers to evaluating a product or service by testing it with representative users. The goal is to identify any usability problems, collect qualitative and quantitative data and determine the participant's satisfaction with the product.”

usability.gov

What is usability?

“If there's one thing you learn by working on a lot of different Web sites, it's that almost any design idea—no matter how appallingly bad—can be made usable in the right circumstances, with enough effort.”

Steve Krug, author
*Don't Make Me Think:
A Common Sense Approach to Web Usability*

Keep in mind

- Popular ≠ Good
- Attractive ≠ Usable
- Ugly ≠ Unusable



What, when,
and who to test?

What to test?

Anything or everything?

What should you test?

- Information Architecture
- Navigation
- User Interface elements
- New Functionality
- Critical User Interactions
- Everything!

When to test?

Is it too soon, or too late?

When to test

- The sooner the better
- Tests should be part of the change process
- Test early and test often!

Who to test?

Your opinion probably isn't the one that matters.

Potential test subjects

- Actual site users
- Representative site users
- Crowdsourced
- Artificial intelligence

Actual site users

Pros:

- Most relevant to site

Cons:

- May have limited availability
- May require compensation

Representative users

Pros:

- Relevant to site
- More relevant than crowdsourced and AI

Cons:

- Only as relevant as your recruiting
- May require compensation

Crowdsourced

Pros:

- Readily available
- Likely more relevant than and AI

Cons:

- May not closely model actual site users
- May require compensation

Artificial intelligence

Pros:

- Readily available
- Complete tests in real time

Cons:

- May not closely model actual human users

Which is right for you?

- What kind of site are you testing?
- What is your site's demographic?
- What access do you have to site users?



How to test?

Types of tests

One size does not fit all!

Formative Tests

- Used throughout the change process.
- Support tool for decision making.
- Different tools for different stages.

Summative Tests

- Generally QA testing.
- Determine if finished feature/site is usable.
- Confirm improved usability.
- Used prior to making changes to determine and document any usability problems.

Types of testing

- Analytics
- User Testing
- Usability Testing

Usability testing

- Based on task completion.
- Usually measurable.

User Testing

- Based on user opinion.
- Not tied to specific task completion.

Analytics

- Measure of actual user behavior.
- May suggest usability problems.
- Not a substitute for usability testing.
- May provide baseline to measure success of plans derived from usability testing.

Analytics

It takes time to collect data; start today!

Web Analytics

Formative

- Records actual site visitor interaction.
- Can suggest usability problems.

Web Analytics

Possible tools

- *Google Analytics* (Google)
- *WebTrends Analytics* (WebTrends)
- *AWStats* (AWStats)

Heat Maps

Formative

- Records specific actual site visitor interaction: mouse click and scroll depth.
- Can suggest usability problems.

Heat Maps

Possible tools

- *Crazy Egg* (Crazy Egg)
- *Hotjar* (Hotjar)
- *Lucky Orange* (Lucky Orange)
- *Google Analytics** (Google)

*sort of

User testing

Opinions are like noses, everyone's got one.

User Surveys

Formative

- Ask users what they like or don't like.
- Good starting point for refining or adding new features.

User Surveys

Possible tools

- Paper
- *SurveyMonkey* (SurveyMonkey)
- *SurveyGizmo* (Widgix)
- *Qualtrics* (Qualtrics)

A/B & Multivariate Tests

Formative

- Users are randomly given an implementation.
- Determine which implementation is most effective based on conversion rates.

A/B & Multivariate Tests

Possible tools

- *Google Website Optimizer* (Google)
- CMS Specific tools (e.g. *Drupal Multivariate*)
- *Optimizely* (Optimizely)

Usability testing

Choose the test based on your question.

Open Card Sort

Formative

- Participants organize a list of items into categories they provide.

Open Card Sort

Possible tools

- Index cards
- *Cardsort* (Optimal Workshop)
- *UserZoom* (UserZoom)

Closed Card Sort

Formative

- Participants organize a list of items into categories you provide.

Closed Card Sort

Possible tools

- Index cards
- *Cardsort* (Optimal Workshop)
- *UserZoom* (UserZoom)

Tree Test

Formative

- Asks test participants where they would go to find specific content.
- Tests the organization of content if navigation labels make sense to users
- Test website's pre-defined navigation.

Tree Test

Possible tools

- Index cards
- *Treejack* (Optimal Workshop)

Expectancy Test

Formative

- Tests a user's impressions of an early prototype of the interface.
- Typically paper prototype, wireframes, or design concepts.

Expectancy Test

Possible tools

- Paper Prototype
- Wireframes
- Design Concepts

5-Second Usability Test

Formative

- Tests individual pages (not homepage) for a specific scenario and task.
- Not suitable for testing complex systems or user interaction.

5-Second Usability Test

Possible tools

- Paper Prototype
- Wireframes
- Design Concepts
- *FiveSecondTest* (UsabilityHub)

Visual Affordance Test

Formative

- Tests the “Clickability/Tappability.”
- Show subject web page or design concept.
- Subject indicate every item believed clickable, and items believed not clickable.

Visual Affordance Test

Possible tools

- Paper Prototype
- Wireframes
- Design Concepts

Click Test

Summative

- Provide scenario and task.
- User clicks on the design element to complete task.

Click Test

Possible tools

- Paper Prototype
- Wireframes
- Design Concepts
- *Chalkmark* (Optimal Workshop)
- *Flippa* (UsabilityHub)

Sample-of-One Test

Summative

- Document fundamental errors in interface layout, labels, content organization, affordance, and functionality
- Use single subject who fits main persona.

Sample-of-One Test

Possible tools

- *Silverback* (Clearleft)
- *OpenHallway* (OpenHallway)
- *TryMyUI* (TryMyUI)

Performance Test

Formative or Summative

- Can be to look for problems in existing site or confirm function in a new site.
- Provide participants a task with a scenario
- Observe user interact with website.

Performance Test

Possible tools

- *Silverback* (Clearleft)
- *OpenHallway* (OpenHallway)
- *TryMyUI* (TryMyUI)

Free Exploration Test

Summative

- Observe users interaction with website for set period of time (5 minutes?).
- Use when navigation is no longer an issue.

Free Exploration Test

Possible tools

- *Silverback* (Clearleft)
- *OpenHallway* (OpenHallway)
- *TryMyUI* (TryMyUI)



Free and Inexpensive Testing Tools

Survey Monkey

Likes:

- Robust question functionality & reporting.
- Filter response data by response or URL.
- Data export to allow manipulation in excel.

Be Aware:

- Some features require more pricey plans.

Survey Monkey

The screenshot shows the SurveyMonkey web application interface. At the top, there's a navigation bar with the SurveyMonkey logo, links for 'My Surveys', 'Examples', 'Survey Services', and 'Plans & Pricing', and a '+ Create Survey' button. A teal banner below the navigation bar says 'Let's stay in touch! Update your email address to get our latest news and updates. [Update now](#)'. The main content area is divided into two sections. The left section, titled 'ALL SURVEYS', contains a table with columns for 'TITLE', 'MODIFIED', 'RESPONSES', and 'ACTIONS'. The table lists six surveys, including 'NISS Affiliate Benefits', 'Carolina Journal Online User Survey', and 'Draft 1 - DH content survey'. The right section, titled 'Need people to take your survey?', promotes 'SurveyMonkey Audience' and includes a 'Buy a Targeted Audience' button. Below this, a 'Get Survey Help, Ideas & Tips' section offers links to 'Learn how to survey like a pro' and 'Find more survey tools to help you'. A 'Feedback' button is visible on the far right.

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Welcome to SurveyMonkey!

SurveyMonkey® Upgrade DesignHammerSurveys

My Surveys Examples Survey Services Plans & Pricing + Create Survey

Let's stay in touch! Update your email address to get our latest news and updates. [Update now](#)

ALL SURVEYS

TITLE	MODIFIED	RESPONSES	ACTIONS
NISS Affiliate Benefits Created 03/13/2015	03/31/2015	28	Edit View Chart
Carolina Journal Online User Survey Created 11/13/2014	01/14/2015	615	Edit View Chart
Draft 1 - DH content survey Created 11/19/2014	11/19/2014	0	Edit View Chart
Carolina Journal User Survey - for JLF internal review Created 11/06/2014	11/11/2014	5	Edit View Chart
Carolina Journal User Survey - reweighted Created 10/31/2014	11/06/2014	10	Edit View Chart
Carolina Journal User Survey test Created 10/27/2014	10/30/2014	0	Edit View Chart

+ Create Survey

ALL SURVEYS: 6 of 6

Need people to take your survey?

Choose from millions of people ready to take your survey with SurveyMonkey Audience.

[Buy a Targeted Audience](#)

Get Survey Help, Ideas & Tips

- [Learn how to survey like a pro](#)
- [Find more survey tools to help you](#)

Feedback

Survey Monkey

The screenshot shows the SurveyMonkey web interface. At the top, there's a navigation bar with the SurveyMonkey logo, links for 'My Surveys', 'Examples', 'Survey Services', and 'Plans & Pricing', and a '+ Create Survey' button. Below this is a yellow banner with the text 'Organize your open ended responses using text analysis. Switch to GOLD →'. The main content area is titled 'Carolina Journal Online User Survey' and has tabs for 'Summary', 'Design Survey', 'Collect Responses', and 'Analyze Results'. The 'Summary' tab is active, showing a 'DESIGN SUMMARY' on the left and a 'RESPONSE SUMMARY' on the right. The 'DESIGN SUMMARY' includes details about the survey (Carolina Journal Online User Survey, Created on 11/13/2014) and a list of features: Questions: 13, Pages: 8; Survey language: English; Theme: Carolina Journal User Survey Oct. 2014; Logo added; Logic added. The 'RESPONSE SUMMARY' shows 615 Total Responses and a 'CLOSED' Overall Survey Status. Below this is a table of 'Collectors' with columns for collector name, responses, start date, and status. A 'Feedback' button is visible on the right side of the page.

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SurveyMonkey Survey Summary - Carolina Journal Online User Survey

SurveyMonkey® Upgrade DesignHammerSurveys + Create Survey

My Surveys Examples Survey Services Plans & Pricing

Organize your open ended responses using text analysis. Switch to GOLD →

Carolina Journal Online User Survey

Summary Design Survey Collect Responses Analyze Results

DESIGN SUMMARY

 **Carolina Journal Online User Survey**
Created on 11/13/2014

- ✓ Questions: 13, Pages: 8
- ✓ Survey language: English
- ✓ Theme: Carolina Journal User Survey Oct. 2014
- ✓ Logo added
- ✓ Logic added

Edit Design Preview Survey

RESPONSE SUMMARY

 SURVEY ALERTS: ON

615
Total Responses

CLOSED
Overall Survey Status

Collectors

 Web Link 1	Responses: 436	Since 11/13/2014	CLOSED
 CJ website popup	Responses: 160	Since 11/20/2014	CLOSED
 CJ Twitter Link	Responses: 4	Since 11/25/2014	CLOSED
 JLF Twitter Link	Responses: 10	Since 11/25/2014	CLOSED
 Facebook: Carolina Journal	Responses: 2	Since 11/25/2014	CLOSED
 Facebook: John...ke Foundation	Responses: 3	Since 11/25/2014	CLOSED

Feedback

Survey Monkey

The screenshot shows the SurveyMonkey Design interface for a survey titled "Carolina Journal Online User Survey". The browser address bar shows "surveymonkey.com". The top navigation bar includes the SurveyMonkey logo, "My Surveys", "Examples", "Survey Services", "Plans & Pricing", and a "+ Create Survey" button. A yellow banner promotes white label surveys with a link to "Upgrade to PLATINUM". The main header shows the survey title and tabs for "Summary", "Design Survey", "Collect Responses", and "Analyze Results". Below the tabs are buttons for "Preview & Test", "Print", and "Next". The left sidebar, labeled "BUILDER", contains sections for "Multiple Choice", "Dropdown", "Matrix / Rating Scale", "Matrix of Dropdown Menus", "QUESTION BANK", "LOGIC", "OPTIONS", and "THEMES". The main content area displays the survey introduction, which includes the "CAROLINA JOURNAL ONLINE" logo and the text: "Thank you for helping us improve the Carolina Journal! Your answers to this survey will remain confidential, and will help us create a better Carolina Journal experience for you. This survey contains 7 questions about the Carolina Journal and how you consume and share news coverage in". The bottom right corner of the interface has a vertical sidebar with "Feedback!", "Get Started", and "Next" buttons.

Survey Monkey

The screenshot shows the SurveyMonkey web interface in a browser window. The browser's address bar shows 'surveymonkey.com'. The page title is 'SurveyMonkey - Collector List'. The SurveyMonkey logo is in the top left, and navigation links for 'My Surveys', 'Examples', 'Survey Services', and 'Plans & Pricing' are in the top center. A 'Create Survey' button is in the top right. A banner below the navigation bar says 'Organize your open ended responses using text analysis. Switch to GOLD →'. The main content area is titled 'Carolina Journal Online User Survey' and has tabs for 'Summary', 'Design Survey', 'Collect Responses', and 'Analyze Results'. Below the tabs is a 'Survey Collectors' section with a '+ New Collector' button. A table lists six collectors, all with a status of 'CLOSED'. The table columns are 'NICKNAME', 'STATUS', 'RESPONSES', and 'DATE MODIFIED'. The collectors are: 'Web Link 1' (436 responses), 'CJ website popup' (160 responses), 'CJ Twitter Link' (4 responses), 'JLF Twitter Link' (10 responses), 'Facebook: Carolina Journal' (2 responses), and 'Facebook: John Locke Foundation' (3 responses). All were created on 11/25/2014 and last modified on Wednesday, January 21, 2015, at 11:19 AM. A 'Feedback' button is on the right side of the page.

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SurveyMonkey - Collector List

SurveyMonkey® Upgrade DesignHammerSurveys

My Surveys Examples Survey Services Plans & Pricing + Create Survey

Organize your open ended responses using text analysis. Switch to GOLD →

Carolina Journal Online User Survey

Summary Design Survey Collect Responses Analyze Results

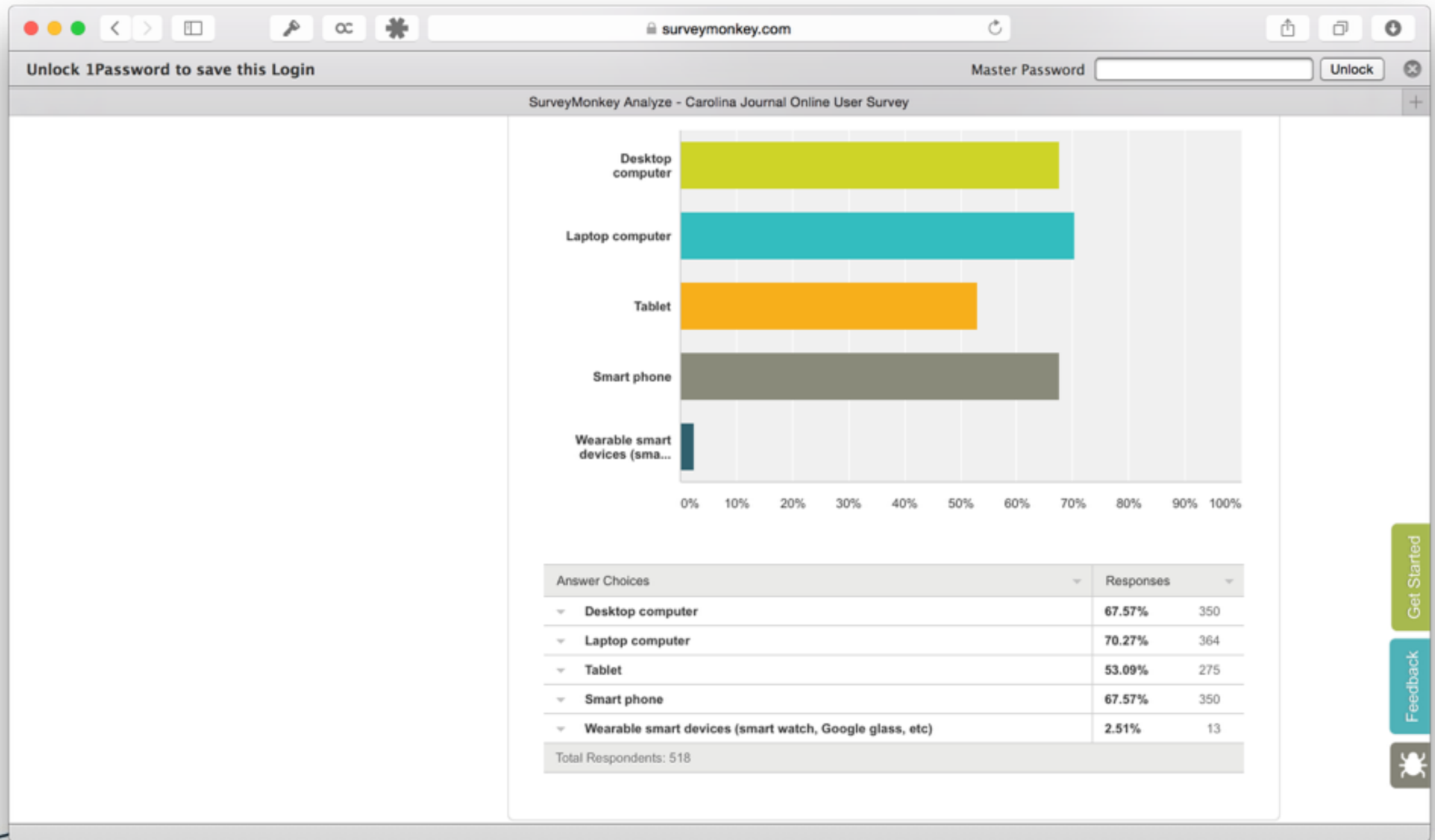
Survey Collectors + New Collector

COLLECTORS: 6 of 6

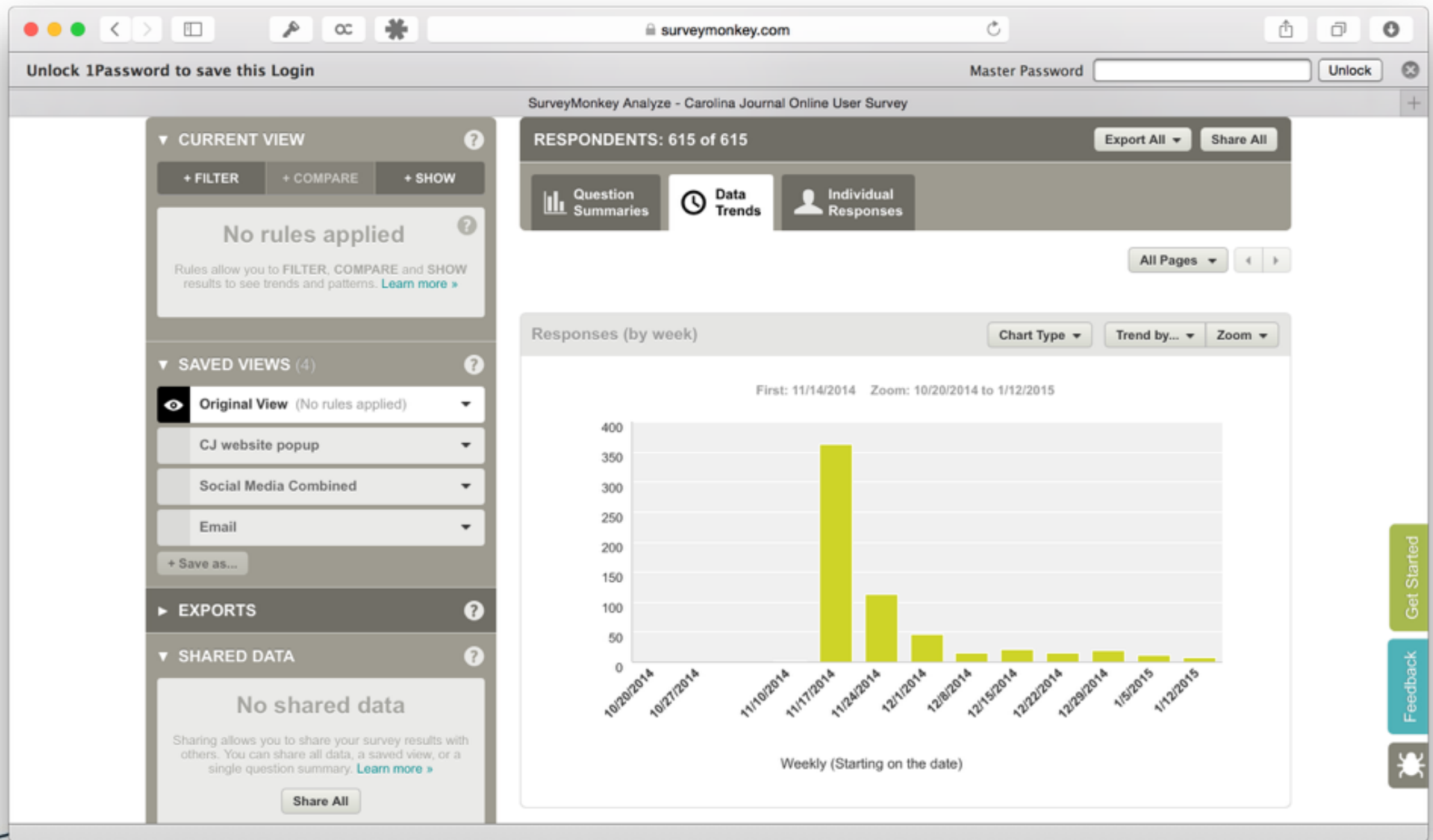
	NICKNAME	STATUS	RESPONSES	DATE MODIFIED	
🔗	Web Link 1 Created 11/13/2014	CLOSED	436	Wednesday, January 21, 2015 11:19 AM	▼
🔗	CJ website popup Created 11/20/2014	CLOSED	160	Wednesday, January 21, 2015 11:19 AM	▼
🔗	CJ Twitter Link Created 11/25/2014	CLOSED	4	Wednesday, January 21, 2015 11:19 AM	▼
🔗	JLF Twitter Link Created 11/25/2014	CLOSED	10	Wednesday, January 21, 2015 11:19 AM	▼
f	Facebook: Carolina Journal Created 11/25/2014	CLOSED	2	Wednesday, January 21, 2015 11:19 AM	▼
f	Facebook: John Locke Foundation Created 11/25/2014	CLOSED	3	Wednesday, January 21, 2015 11:19 AM	▼

Feedback

Survey Monkey



Survey Monkey



Survey Monkey

The screenshot displays the SurveyMonkey web interface in a browser window. The address bar shows 'surveymonkey.com'. The page title is 'SurveyMonkey Analyze - Carolina Journal Online User Survey'. The SurveyMonkey logo is at the top left, with navigation links for 'My Surveys', 'Examples', 'Survey Services', and 'Plans & Pricing'. A 'Create Survey' button is on the right. A banner below the navigation bar offers a 25% discount on an annual plan. The main content area is titled 'Carolina Journal Online User Survey' and includes tabs for 'Summary', 'Design Survey', 'Collect Responses', and 'Analyze Results'. The 'Analyze Results' tab is active, showing 'RESPONDENTS: 615 of 615'. On the left, a 'CURRENT VIEW' sidebar shows 'No rules applied' and a list of 'SAVED VIEWS' including 'Original View', 'CJ website popup', 'Social Media Combined', and 'Email'. The main panel displays 'Individual Responses' for 'Respondent #615', who is 'COMPLETE'. The response details include: Collector: CJ website popup (Web Link), Started: Wednesday, January 14, 2015 3:32:29 PM, Last Modified: Wednesday, January 14, 2015 3:36:14 PM, Time Spent: 00:03:44, and IP Address: 174.109.223.134. Below this, 'PAGE 2' of the survey is shown, with the question: 'Q1: Within the past 12 months, have you visited the Carolina Journal website, or are subscribed to CJ emails?'. A 'Get Started' button is visible on the right side of the interface.

CrazyEgg

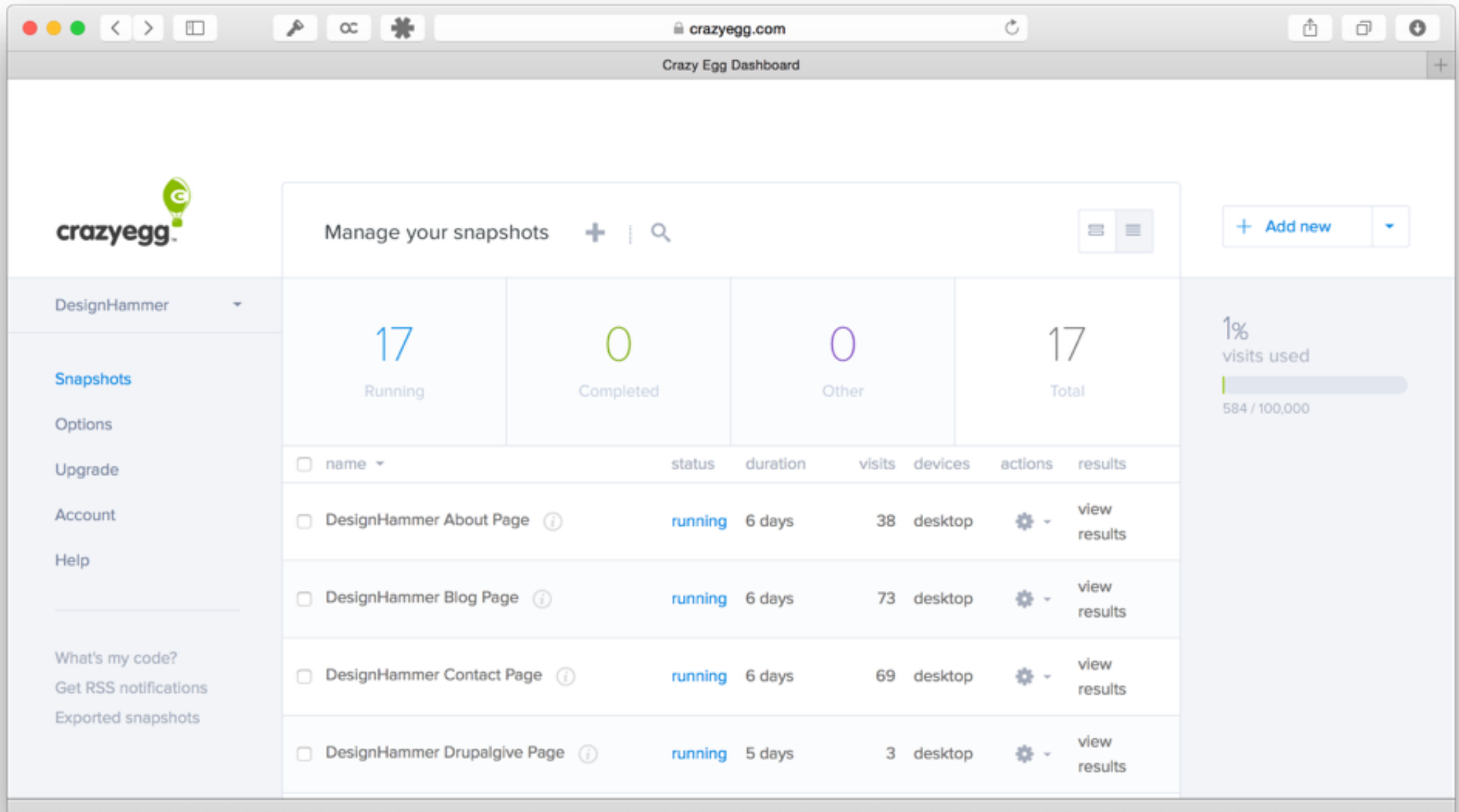
Likes:

- Variety of ways to filter/visual data
- Can test mobile and desktop

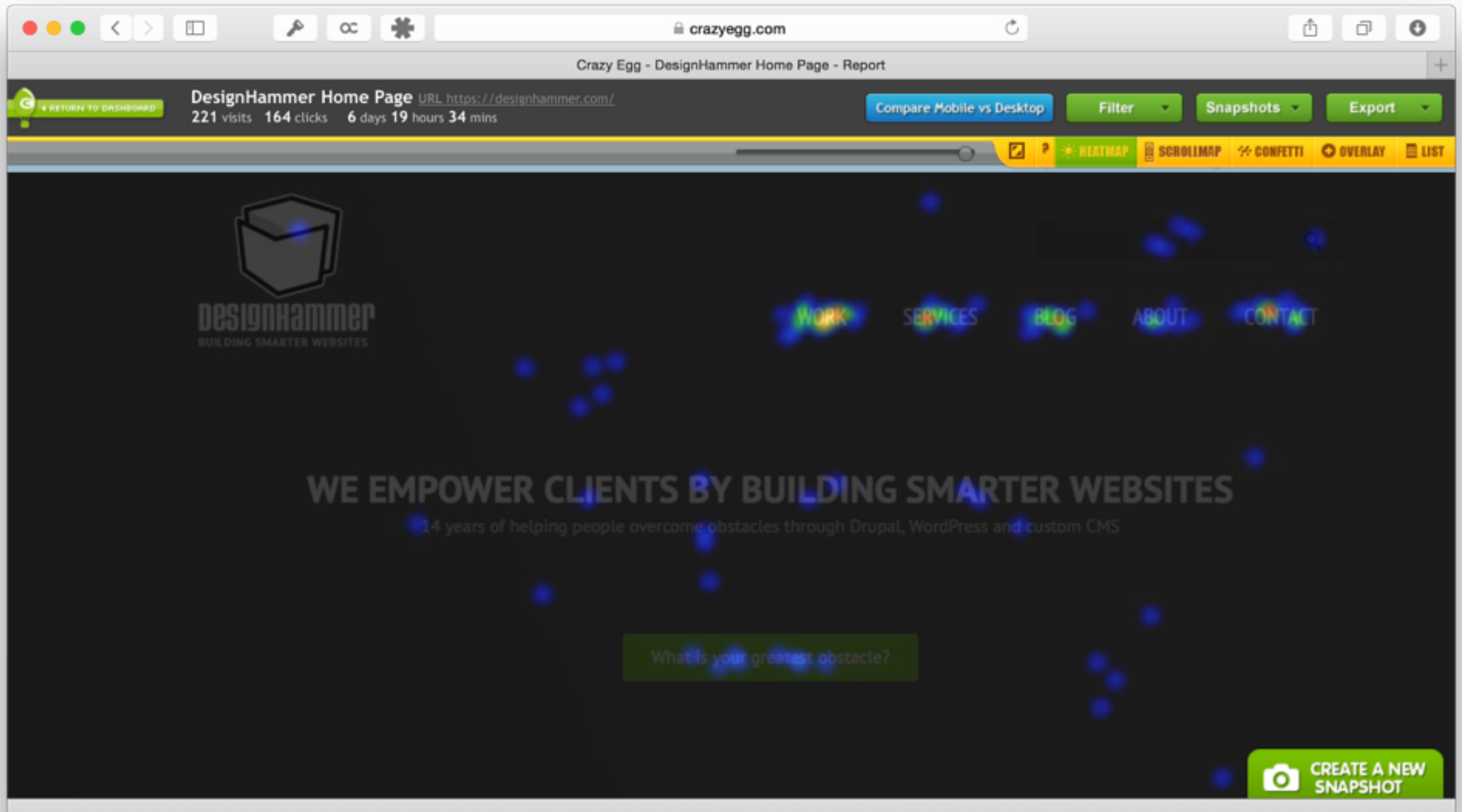
Be Aware:

- Not real time.
- Not cheap to monitor entire sites.

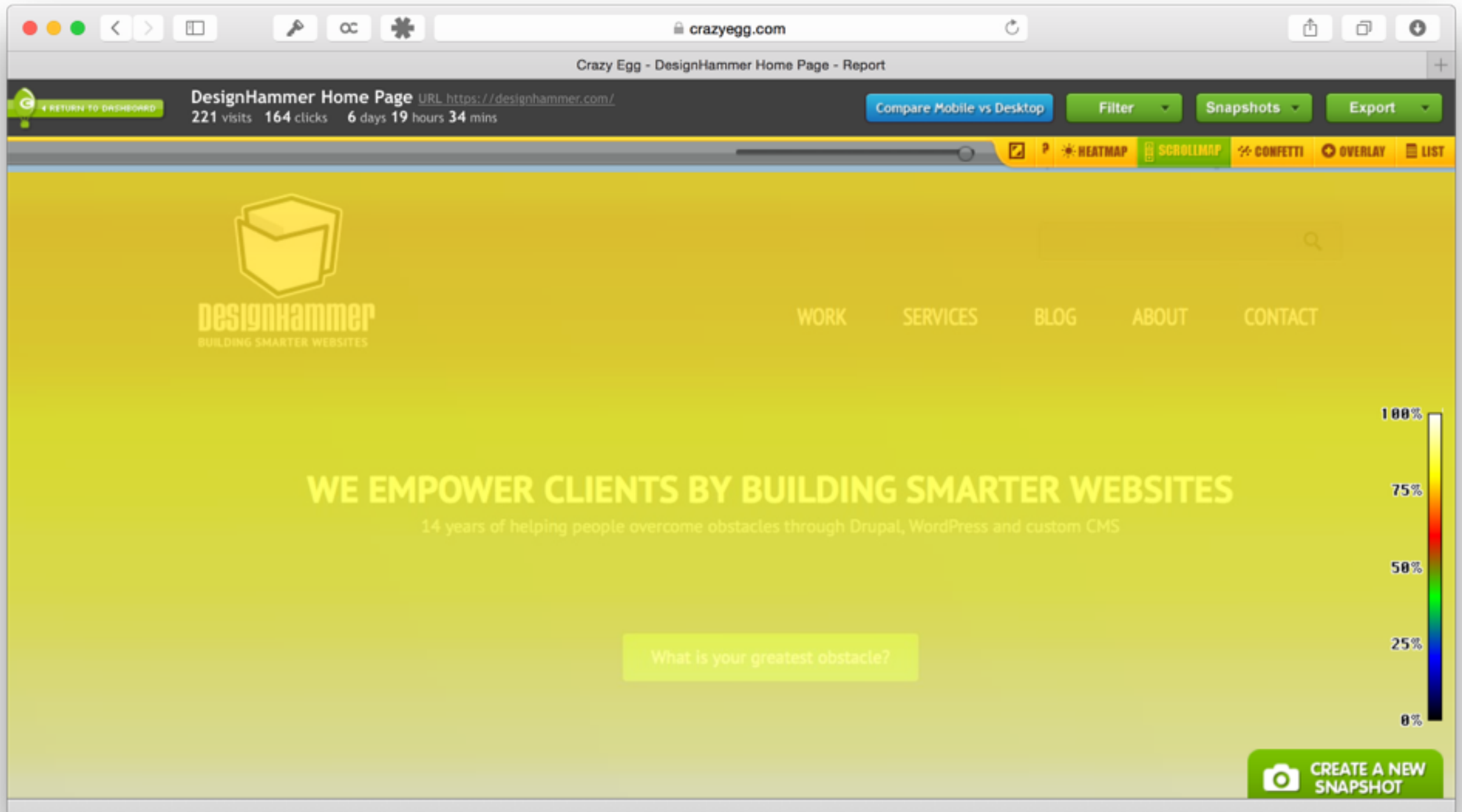
Crazy Egg



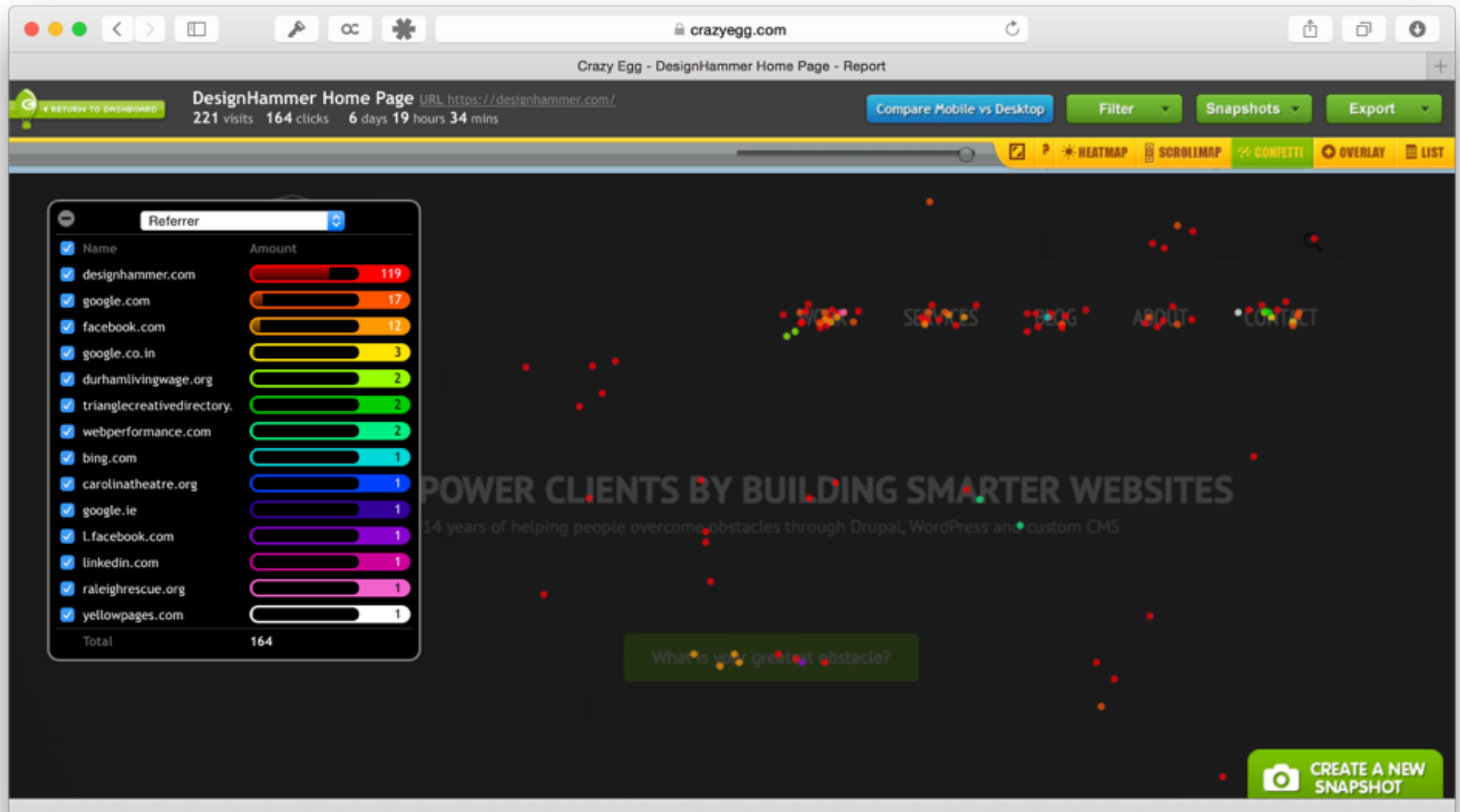
Crazy Egg



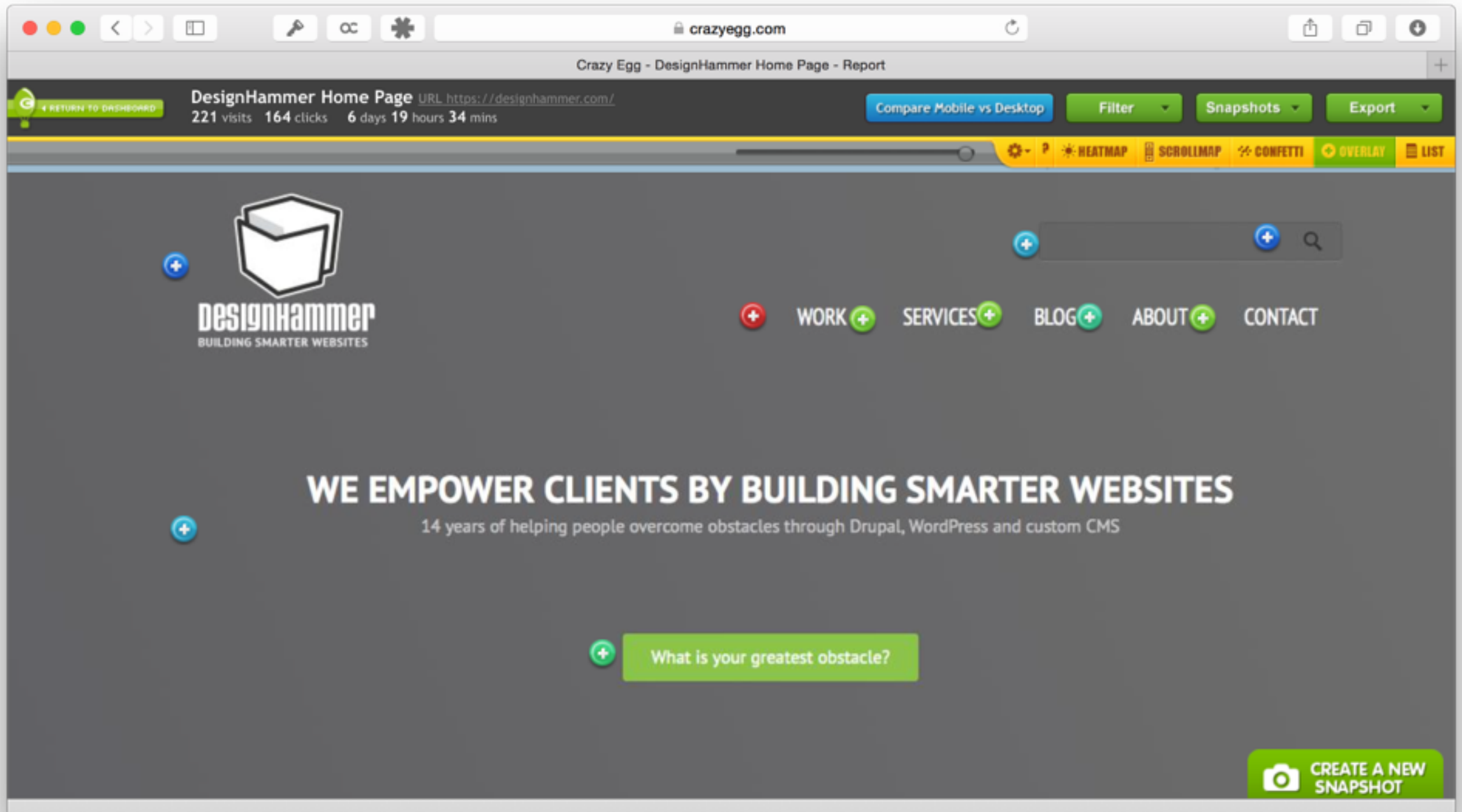
Crazy Egg



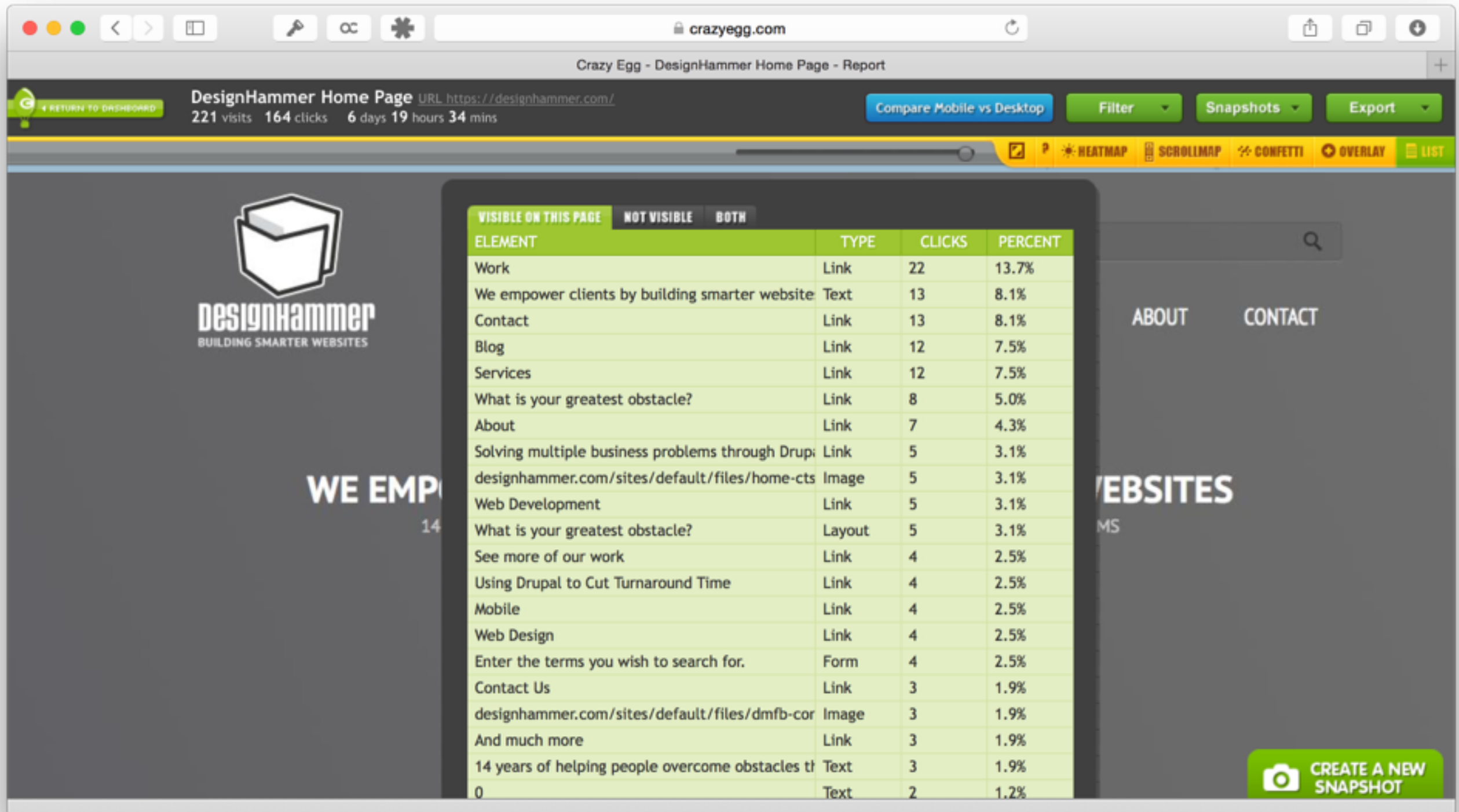
Crazy Egg



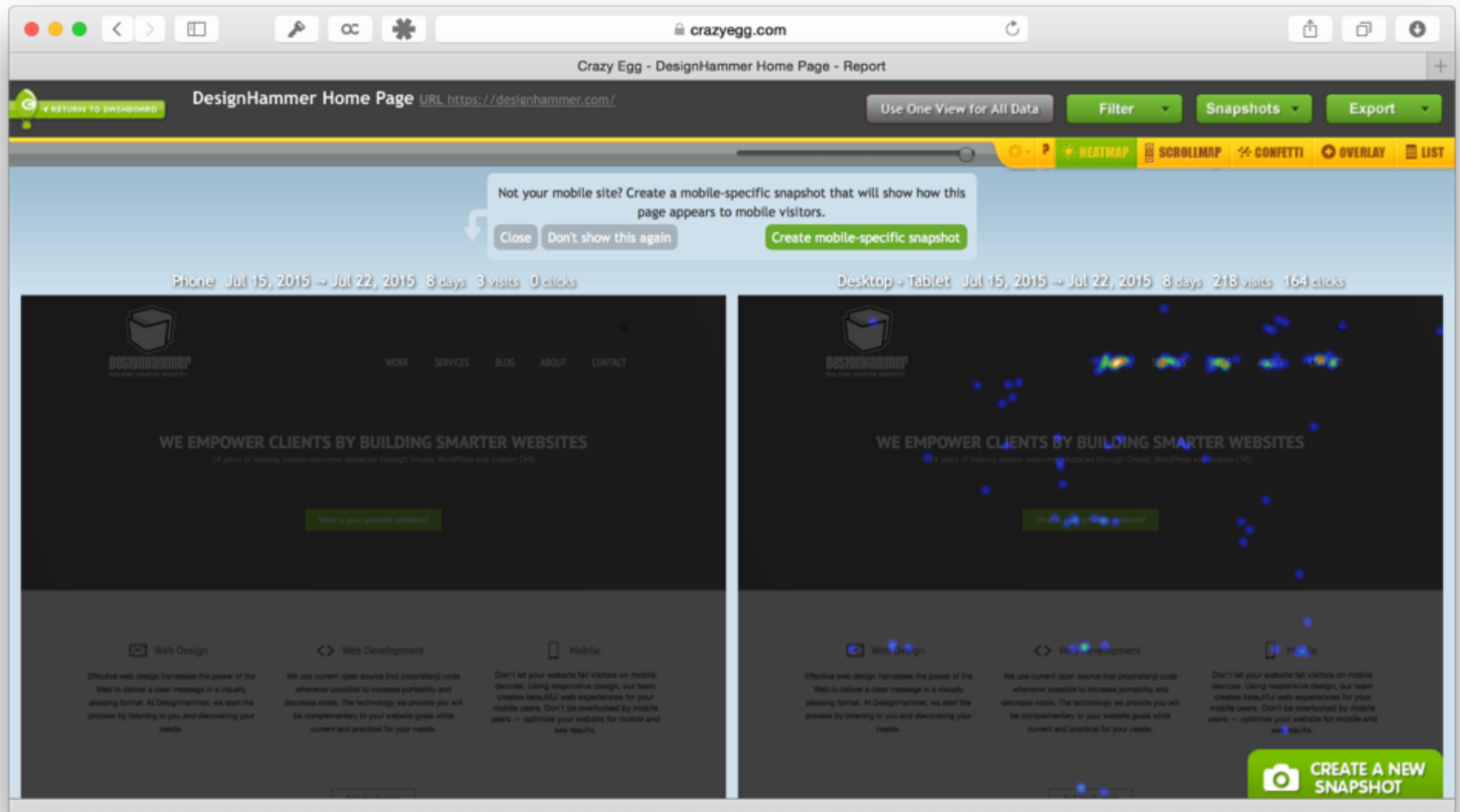
Crazy Egg



Crazy Egg



Crazy Egg



Optimal Sort

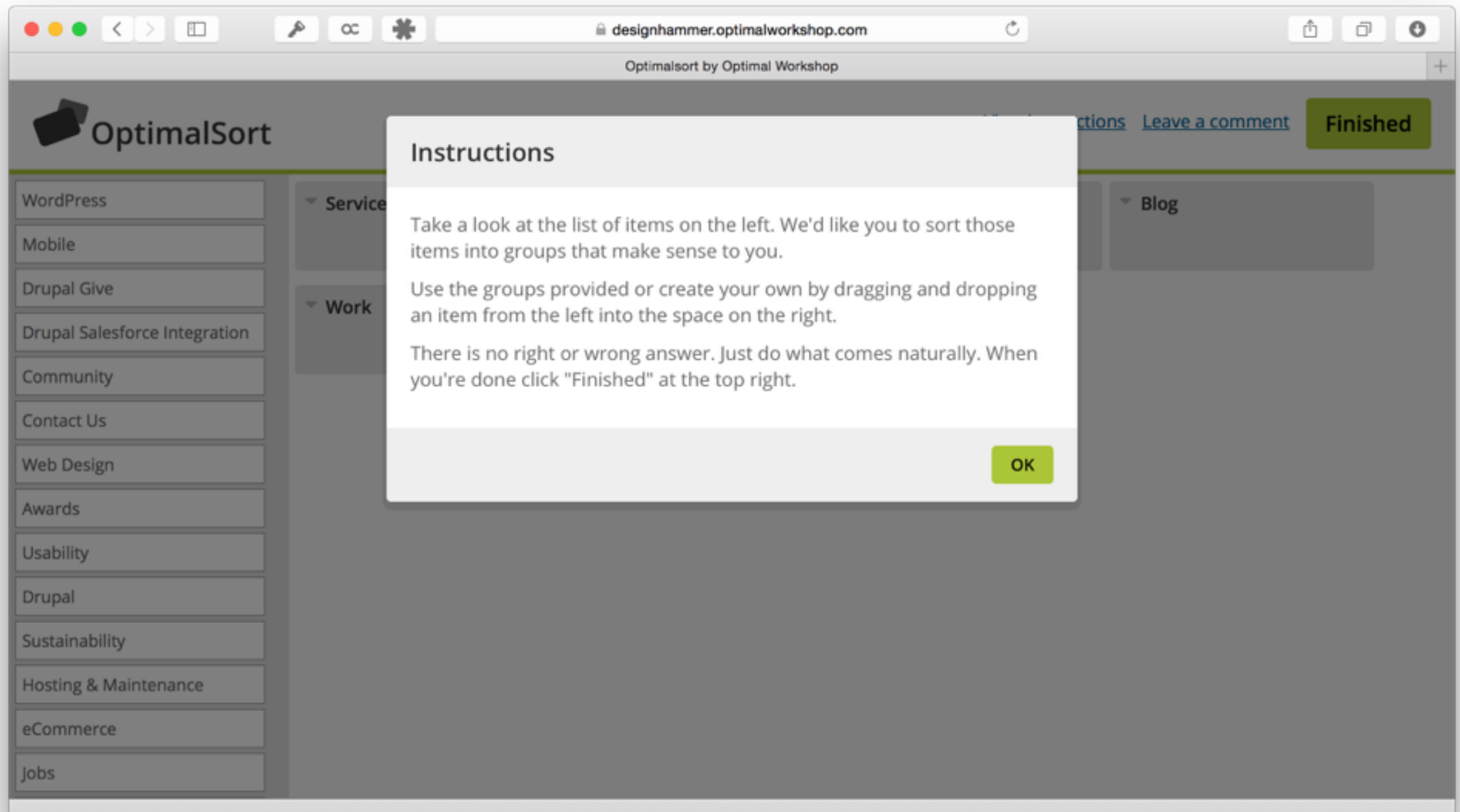
Likes:

- Easy to set up.
- Can use for remote testing.

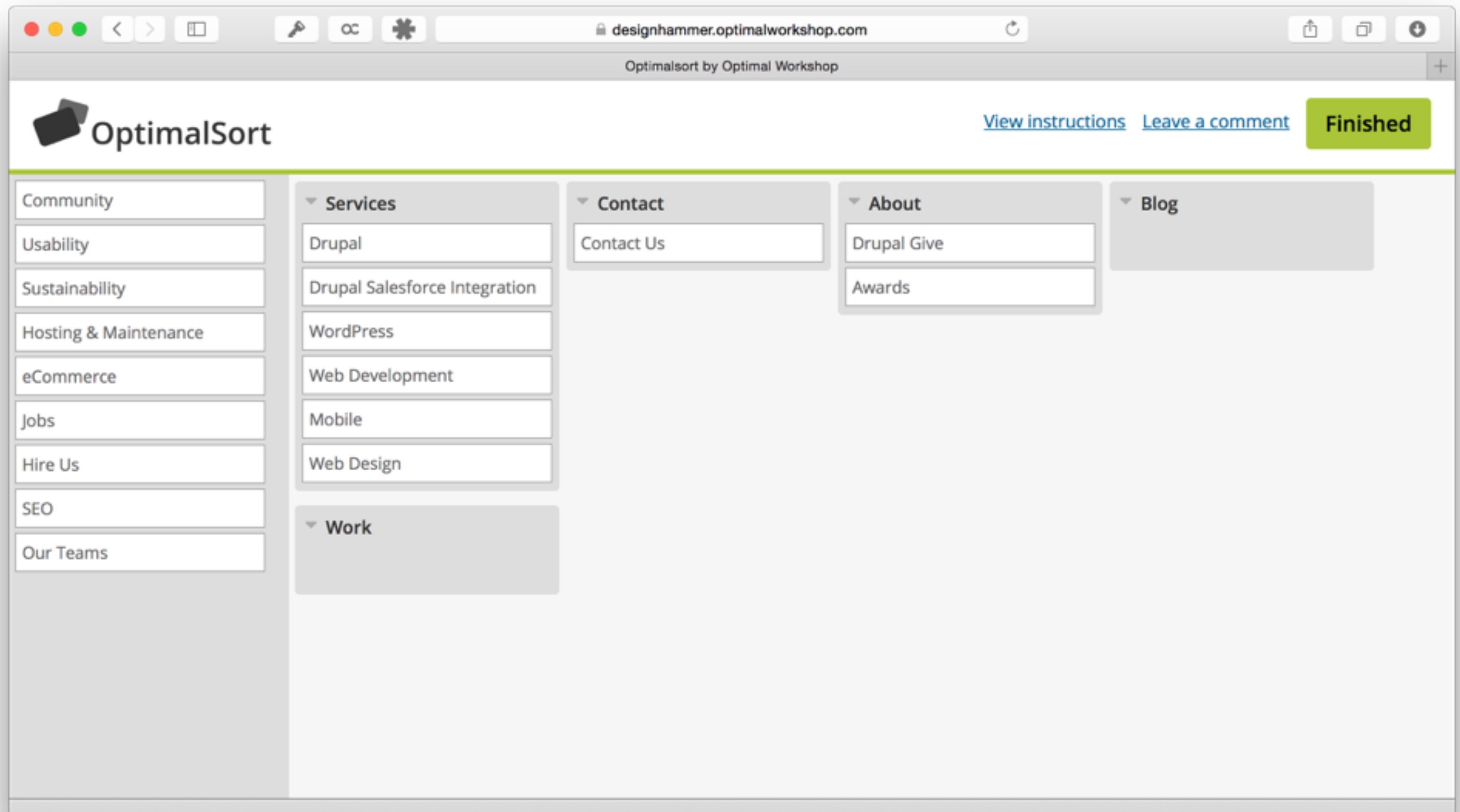
Be Aware:

- Normalizing data requires assumptions.
- Not cheap if you need the full version.

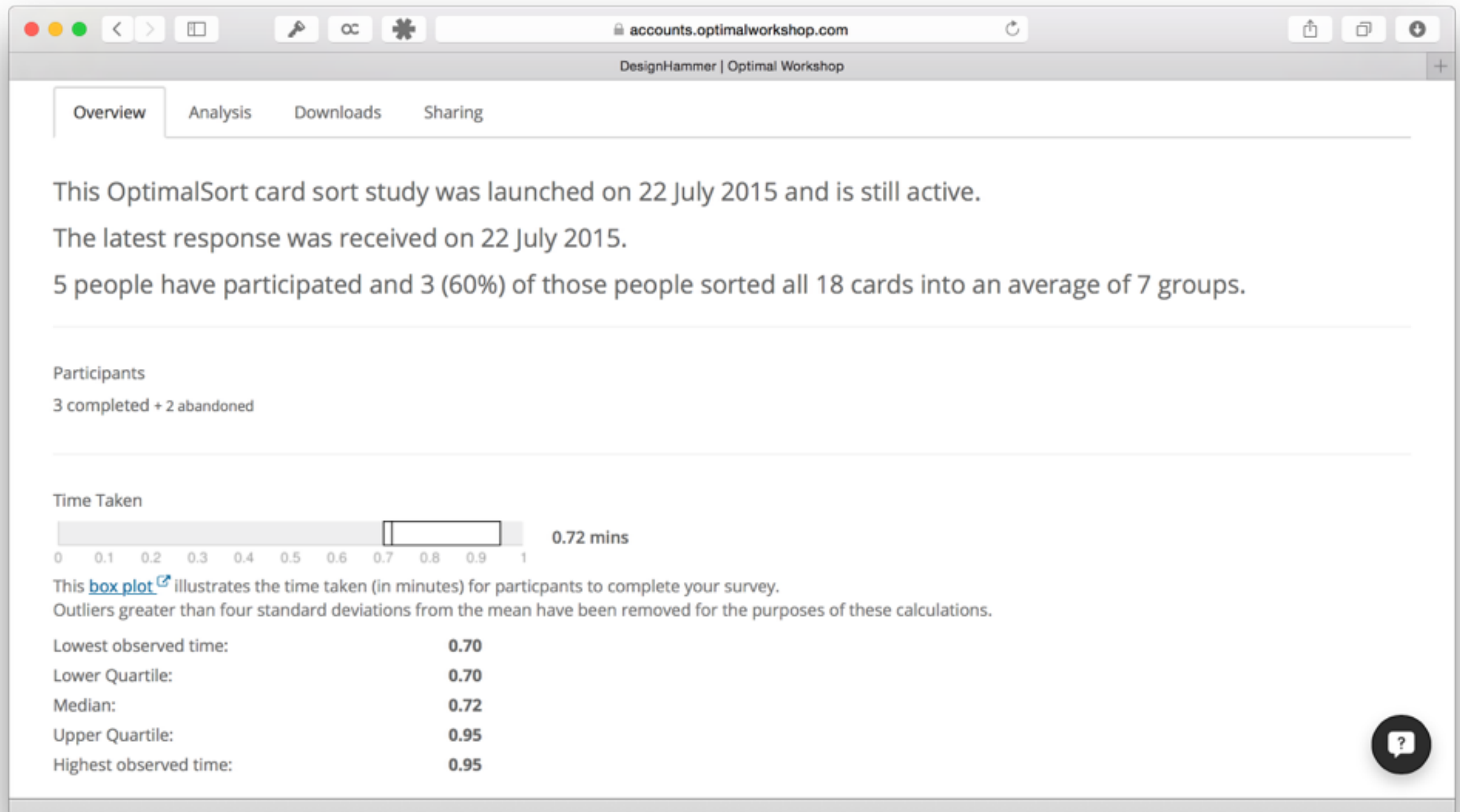
Optimal Sort



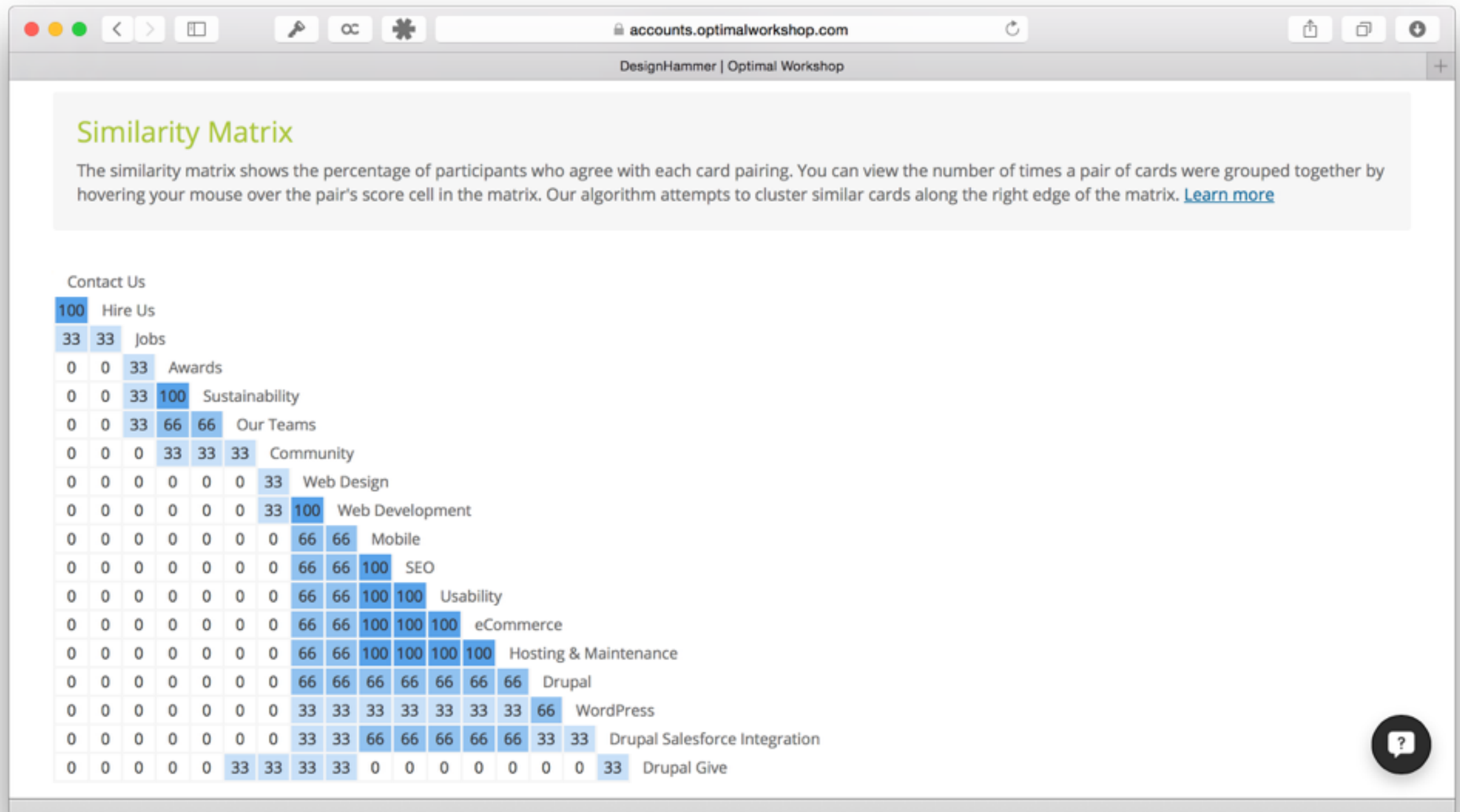
Optimal Sort



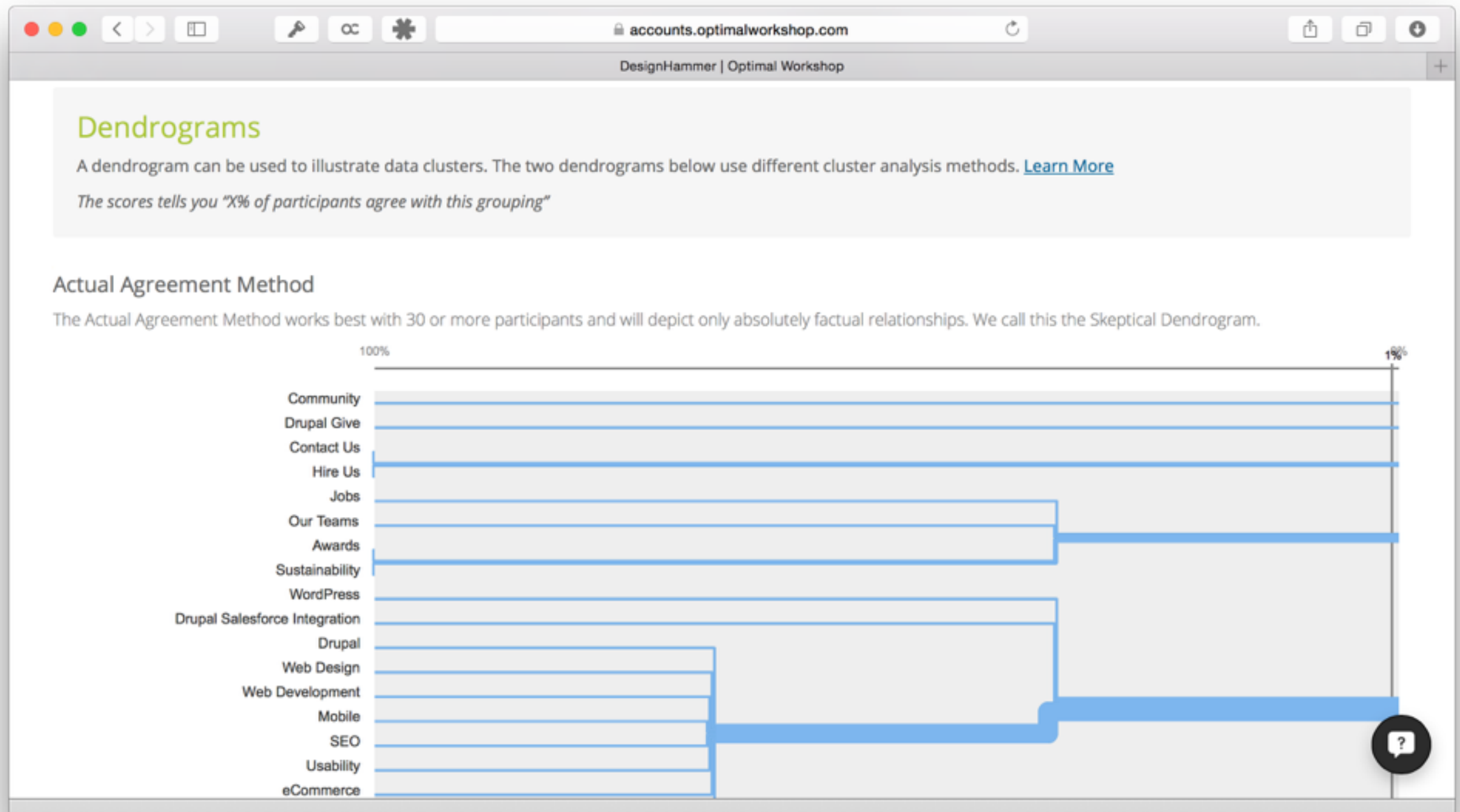
Optimal Sort



Optimal Sort



Optimal Sort



Optimal Sort

The screenshot shows the Optimal Workshop interface. The browser address bar displays 'accounts.optimalworkshop.com'. The navigation bar includes 'Overview', 'Analysis' (selected), 'Downloads', and 'Sharing'. Below this, a sub-navigation bar lists 'Participants', 'Cards', 'Categories', 'Standardization Grid', 'Similarity Matrix', 'Dendrograms', and 'PCA' (selected). The main content area is titled 'Participant Centric Analysis' in green. A descriptive text states: 'The Participant-Centric Analysis shows the top three most acceptable IA submissions by participants in this card sort, as tested against all other participant IAs.' A slider for 'Consider IAs with' is set to '2 - 7 groups'. Three analysis results are displayed as cards:

- Similar IAs: 2/3**
6 groups submitted by stephen@pashby.net
Contact
Similar group labels
Contact Us
Hire Us
Services
Similar group labels
- Similar IAs: 1/3**
5 groups submitted by stephen@designhammer.com
Contact
Similar group labels
Contact Us
Hire Us
Blog
- Similar IAs: 2/3**
5 groups submitted by stephen@pashby.org
Services
Similar group labels
WordPress
Drupal
Web Design
Web Development

A help icon (question mark in a circle) is visible in the bottom right corner of the results area.

Treejack

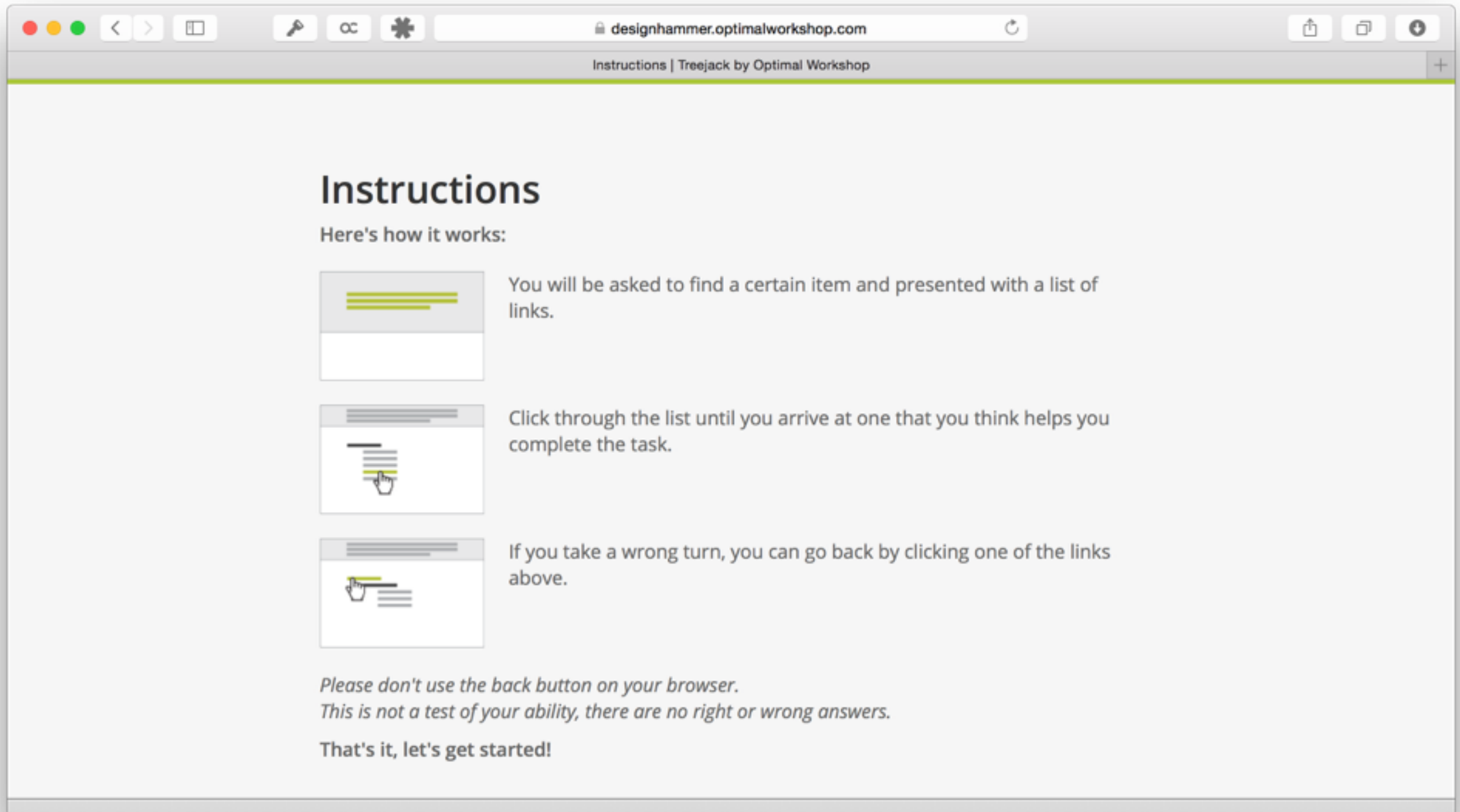
Likes:

- Easy to set up.
- Can use for remote testing.

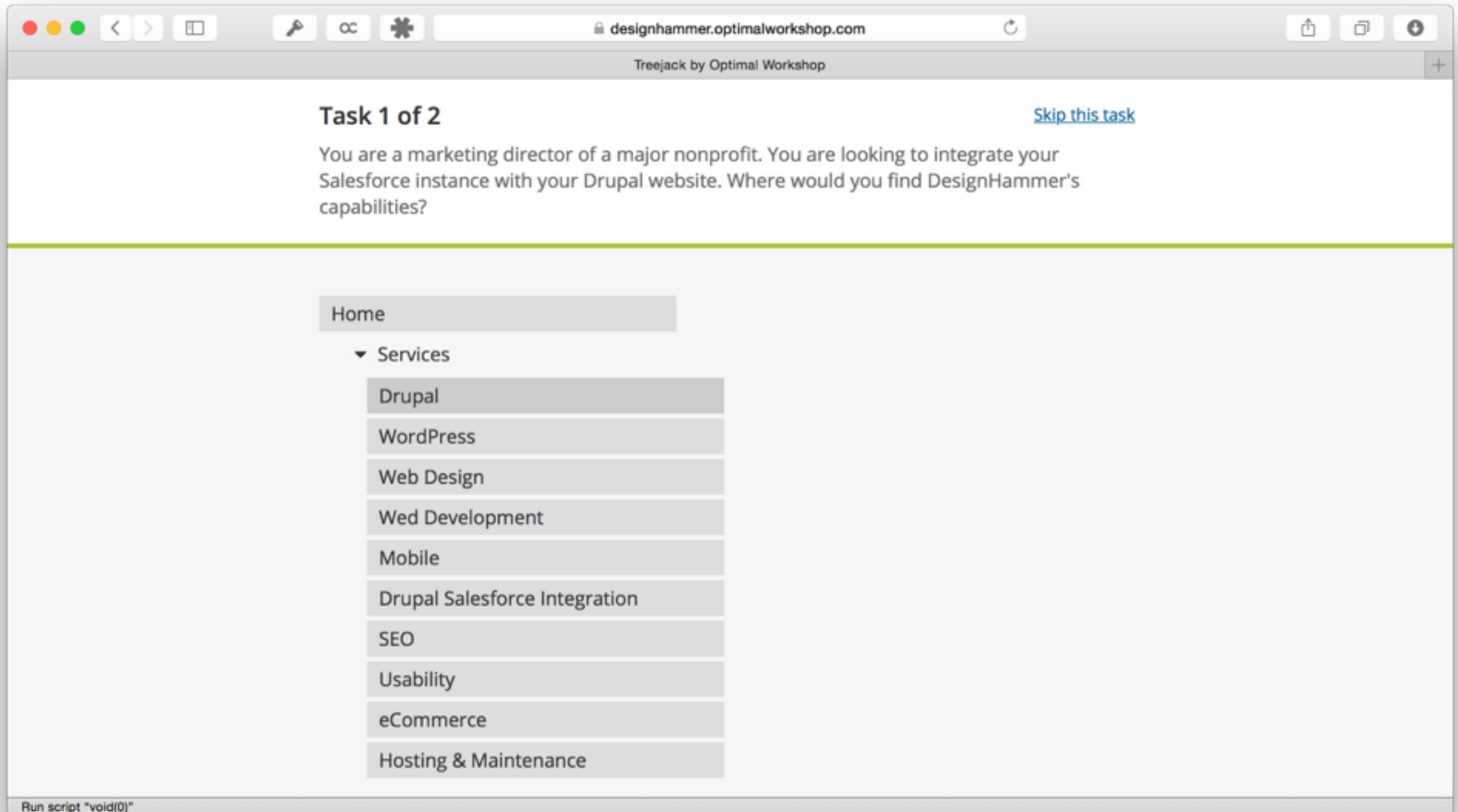
Be Aware:

- Not cheap if you need the full version.

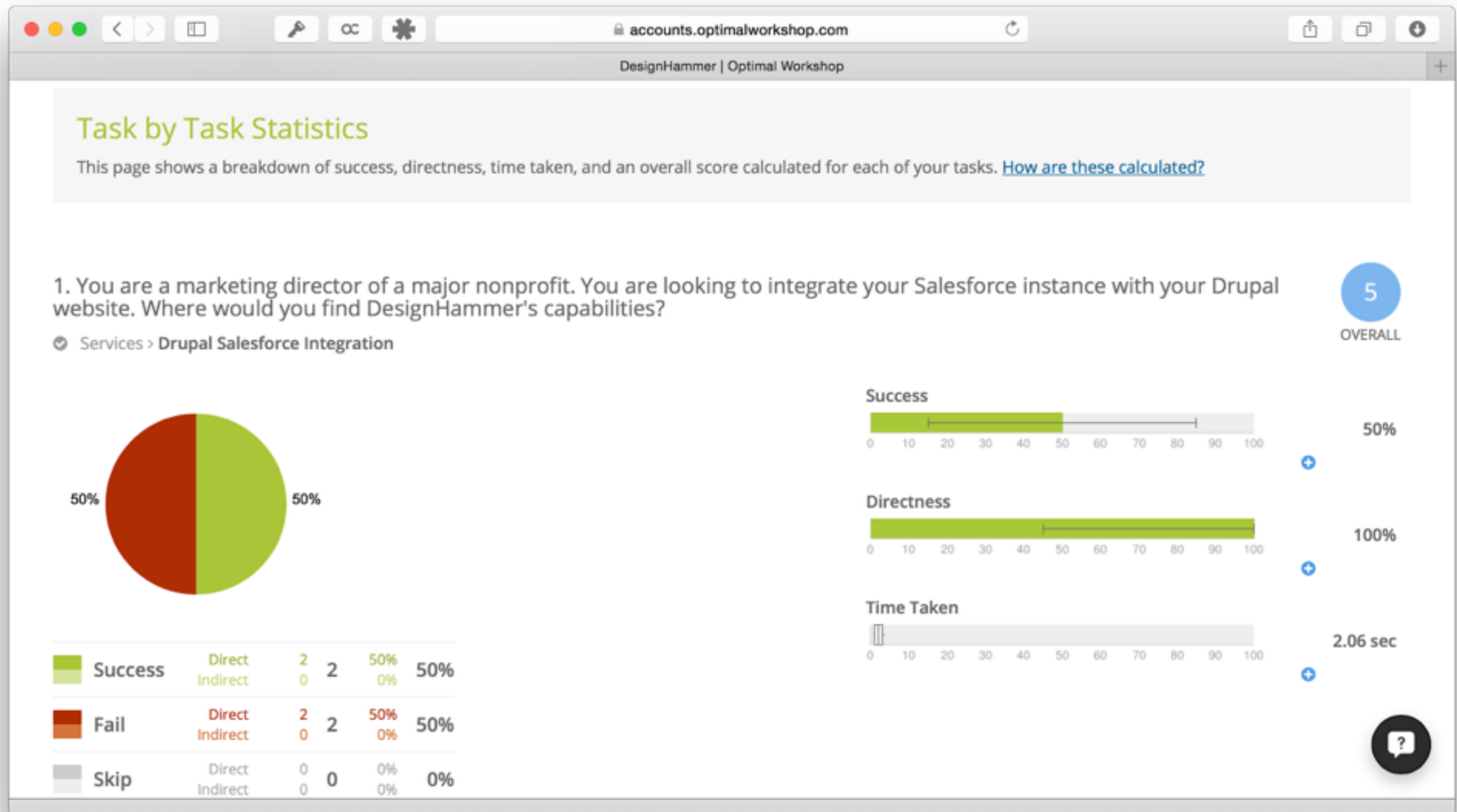
Treejack



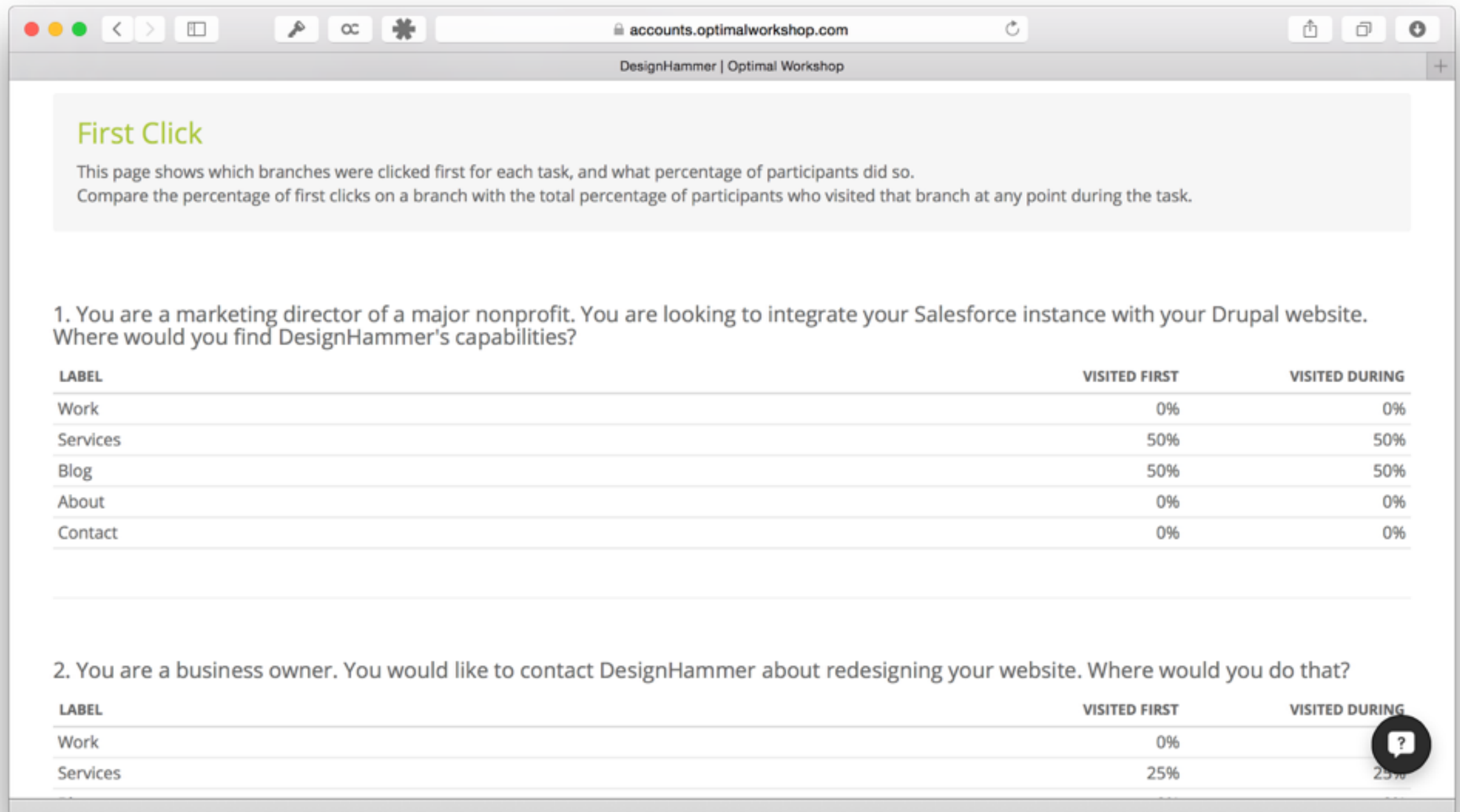
Treejack



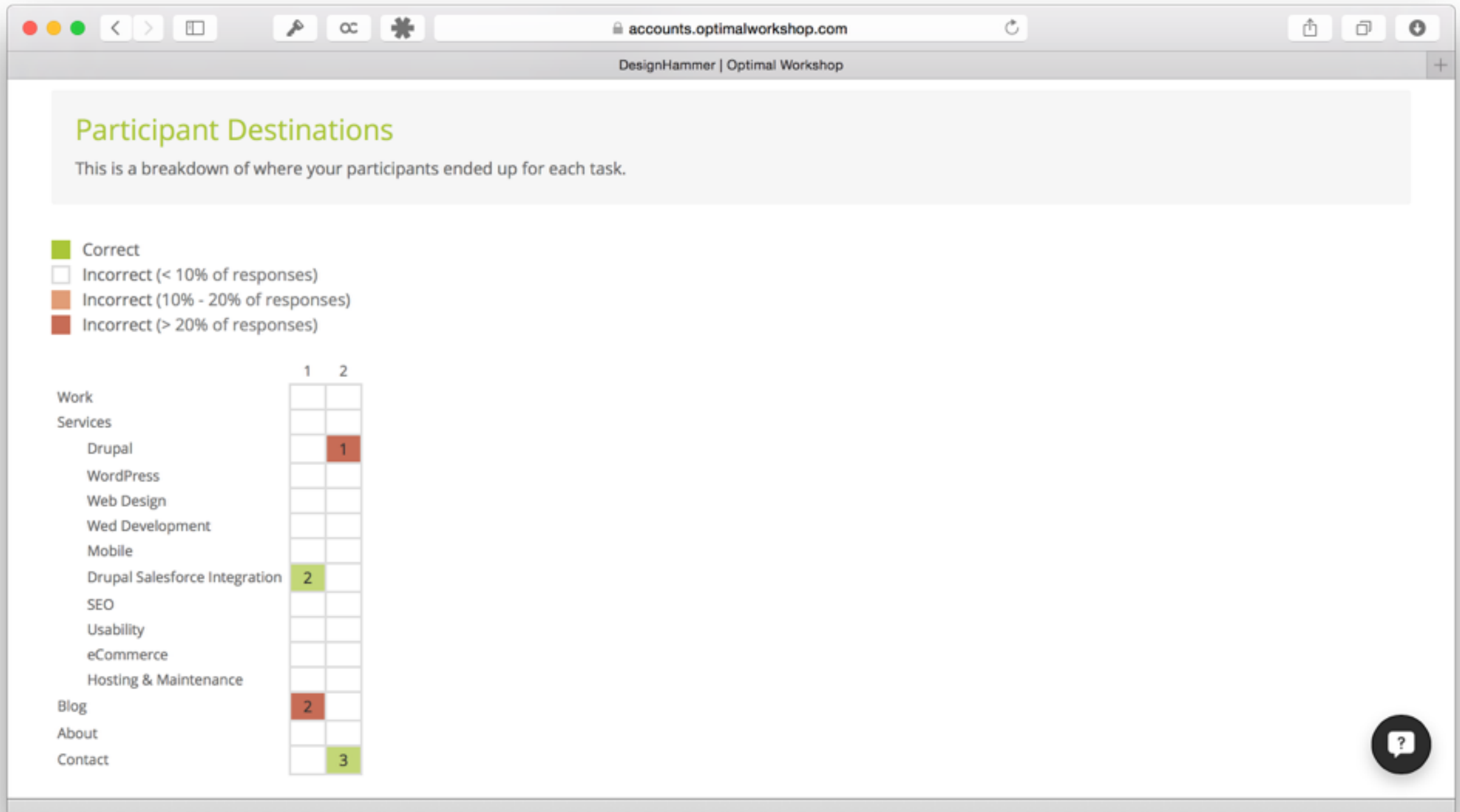
Treejack



Treejack



Treejack



Chalkmark

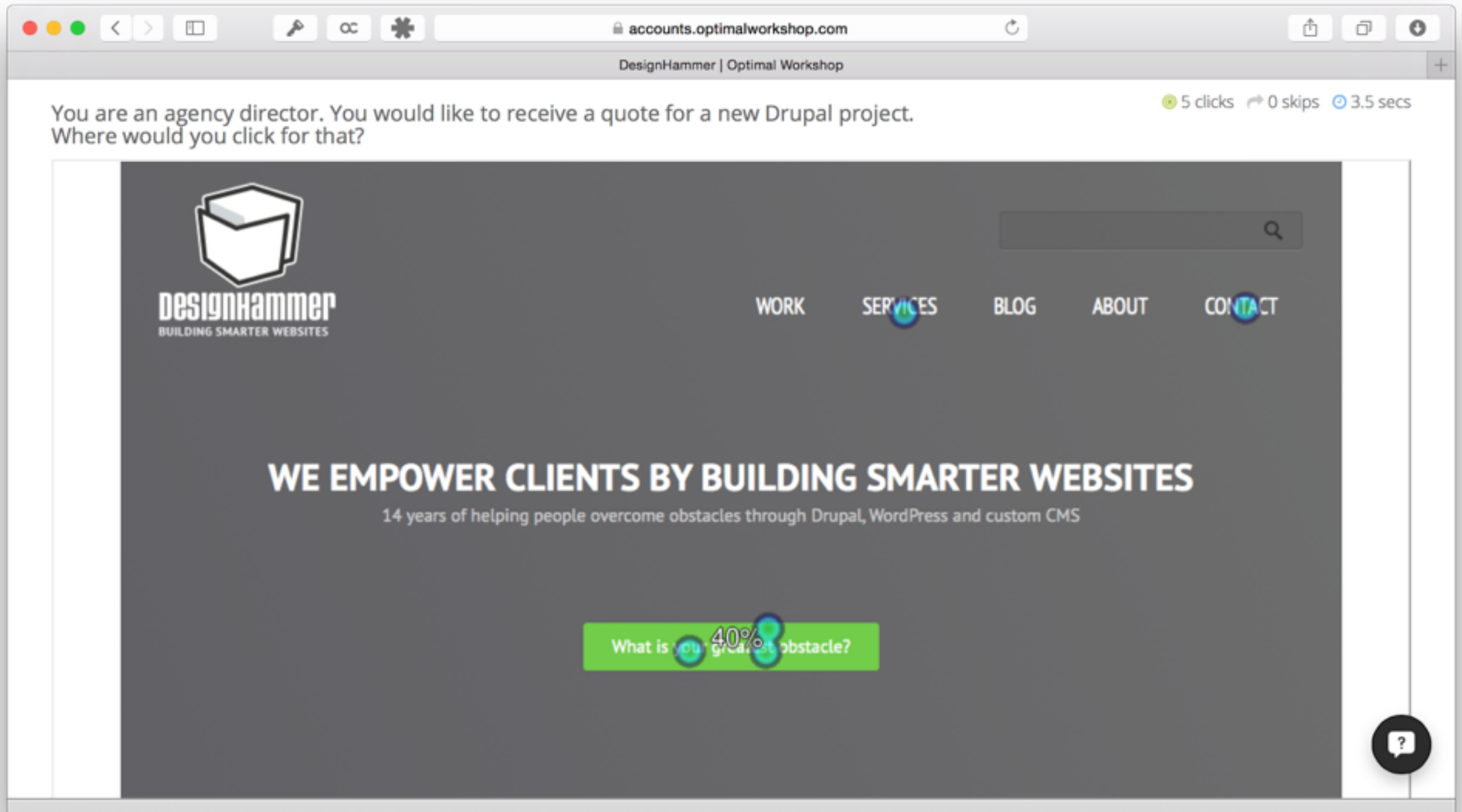
Likes:

- Easy to set up.
- Can use for remote testing.

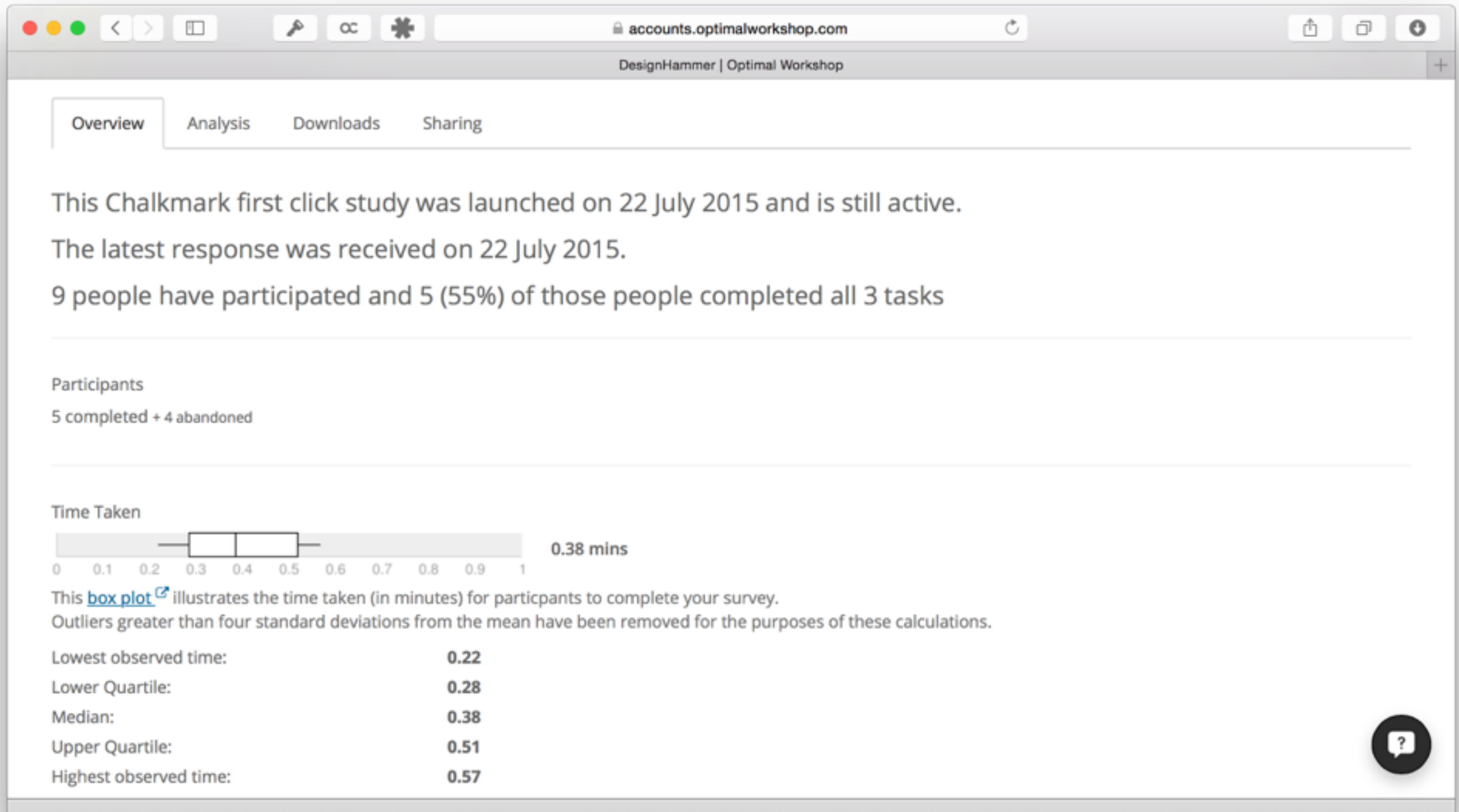
Be Aware:

- Only single response per question.
- Not cheap if you need the full version.

Chalkmark



Chalkmark



Chalkmark

The screenshot shows a web browser window with the URL `accounts.optimalworkshop.com`. The page title is "DesignHammer | Optimal Workshop". The navigation bar includes tabs for "Overview", "Analysis", "Downloads", and "Sharing". Under the "Analysis" tab, there are sub-tabs for "Participants" and "Clickmaps". The "Participants" sub-tab is active, showing a section titled "Your Participants" with a description: "This table lists all of the participants that started your survey. Incomplete responses are marked as 'Abandoned'. Use this table to segment your data, filter responses and reload your results." Below this is a table with columns: PARTICIPANT, IDENTIFIER, TIME TAKEN, QUESTIONS, TASKS COMPLETED, TASKS SKIPPED, and INCLUDE. The table lists five participants, all with 100% tasks completed. A sixth participant is marked as "Abandoned" with 0% tasks completed. A search bar and a "Reload results using current selection" button are located above the table. A help icon (?) is visible in the bottom right corner.

PARTICIPANT	IDENTIFIER	TIME TAKEN	QUESTIONS	TASKS COMPLETED	TASKS SKIPPED	INCLUDE
1	stephen@designhammer.com	00:00:23		100%	0%	☒
2	stephen@pashby.org	00:00:13		100%	0%	☒
3	stephen@pashby.org	00:00:34		100%	0%	☒
4	david@designhammer.com	00:00:21		100%	0%	☒
5	stephen@example.com	00:00:27		100%	0%	☒
6	stephen@pashby.org	Abandoned		0%	0%	☐

Chalkmark

The screenshot shows a web browser window with the address bar displaying `accounts.optimalworkshop.com`. The browser's address bar also shows the page title "DesignHammer | Optimal Workshop". The main navigation bar is green and contains the following items: "Optimal Workshop" (with a logo), "Treejack", "OptimalSort", "Chalkmark" (highlighted), "Reframer", and a user profile "David Minton" with a dropdown arrow. Below the navigation bar, the page title "DesignHammer" is displayed next to a green icon, with an "Edit" button to its right. A tabbed interface below the title shows "Overview", "Analysis", "Downloads" (selected), and "Sharing". The "Downloads" section has a heading "Downloads" and a description: "To perform fine-grained analysis, you may want to download the raw data for your entire survey and manipulate it in a spreadsheet or some other tool." Below this, there are three download options, each with a description and two buttons: "Uncut" and "Filtered".

Download Option	Description	Buttons
Heatmaps (.pdf)	This pdf table contains the heatmaps for each tasks for this survey.	Uncut, Filtered
Participant Data (.xlsx)	This spreadsheet contains the raw data collected about your participants.	Uncut, Filtered
Raw Data (.xlsx)	This spreadsheet contains all of the raw data collected for this survey.	Uncut, Filtered

Silverback

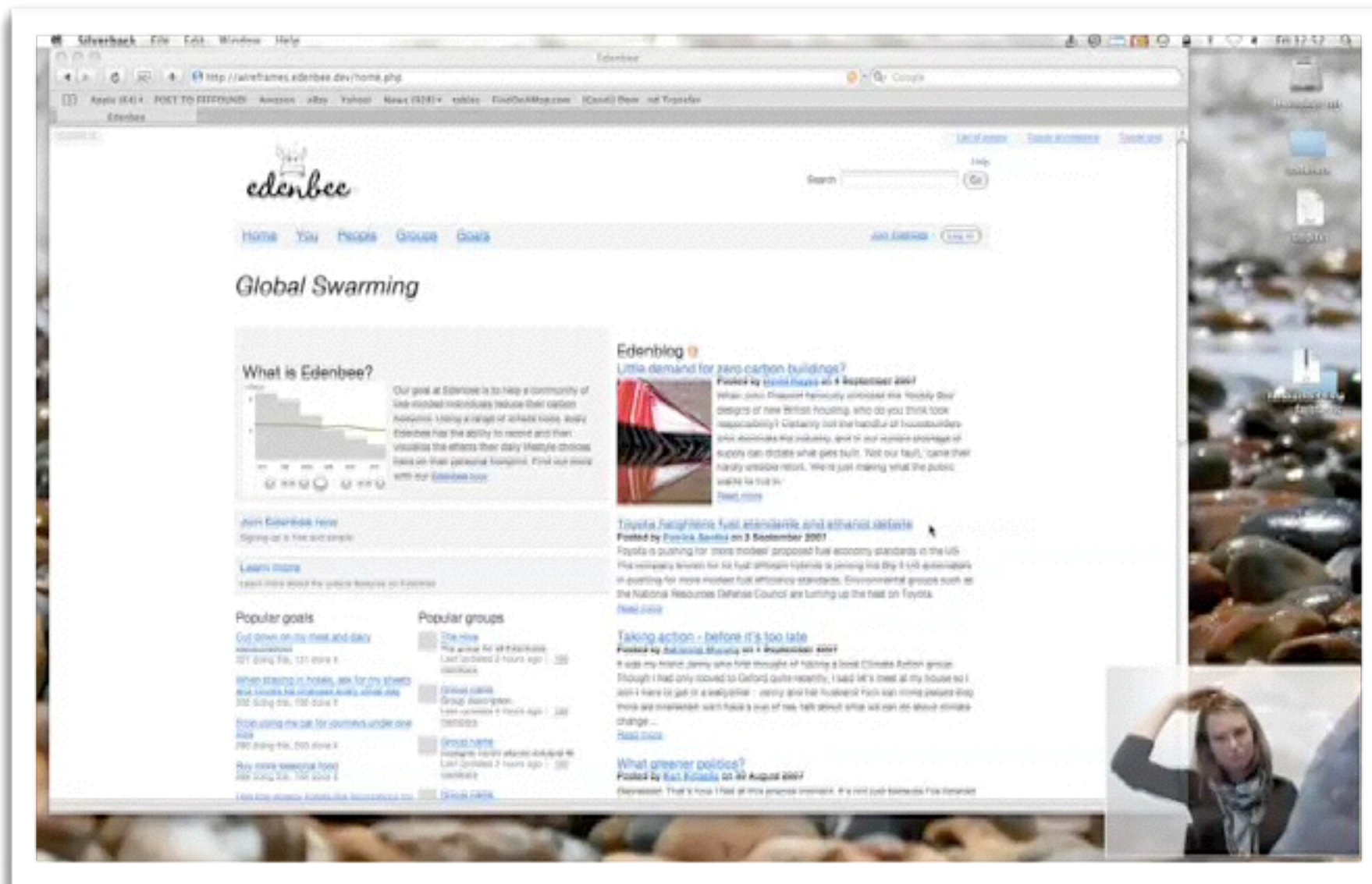
Likes:

- Portable, works well on a standard laptop.
- Does not require a dedicated usability lab.

Be Aware:

- Mac only.
- For on location testing.

Silverback





Conclusion

Conclusion

- No silver bullet to make sites usable.
- Usability is a never-ending process.
- Making sites more usable is fun for all!

Special thanks





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